

TARUN REDDY THANDU

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PROFESSIONAL SUMMARY

AI Engineer specializing in **autonomous agents** and **AI security**. Proven ability in designing complex **multi-agent systems** using **RAG**, **MCP**, and **fine-tuned LLMs**. Experienced in **AI Red Teaming** and deploying scalable services on **AWS**.

TECHNICAL SKILLS

AI & ML : PyTorch, TensorFlow, Python, SQL, FastAPI, Transformers, Fine-tuning, RLHF, DPO
Agents & LLMs : LangGraph, LangChain, CrewAI, Autogen, LangSmith, Langfuse, LiteLLM, OpenTelemetry, MCP
Data & Databases : AWS (S3, RDS), Pinecone, FAISS, Neo4j, ClickHouse, PostgreSQL, MongoDB, Redis, Kafka, ETL
Cloud & MLOps : AWS (SageMaker, Fargate, Bedrock, EKS, Glue), Docker, Kubernetes, Kubeflow, MLflow, W&B

EXPERIENCE

PointGuard AI | AI Engineer - Enterprise security Aug 2025 - Present
LangGraph, MCP, LangFuse, LiteLLM, OpenTelemetry, ClickHouse, Autogen, CrewAI, Docker, FastAPI San Francisco, CA

- Led AI Red Teaming by designing multi-agent workflows with enterprise MCP servers using **LangGraph**, **Autogen**, and **CrewAI** to simulate sophisticated attacks against Fortune 500 client systems to identify critical vulnerabilities such as prompt injections.
- Engineered a novel **Guardian Agent** framework to provide real-time threat detection and remediation. This system monitors all LLM calls and tool invocations to automatically block OWASP LLM Top 10 attacks and alert security teams.
- Architected a production-grade observability pipeline using Langfuse, LiteLLM, and **OpenTelemetry** to capture and analyze agent traces in **ClickHouse**, providing the high-resolution data needed to identify complex attack patterns and security loopholes.
- Developed custom detection models by performing extensive **adversarial testing** (e.g., jailbreaks, content concretization) and trained LLMs to recognize and flag malicious agent behavior patterns with high accuracy.

HER Heard | AI Engineer Intern - Healthcare Sep 2024 - Dec 2024
LangGraph, RAG, QLoRA, RLHF, Pinecone, Neo4j, AWS (Bedrock, SageMaker, Fargate, S3), Kafka, Redis, FastAPI Boston, MA

- Architected a stateful multi-agent medical information chatbot using **LangGraph**, orchestrating specialized agents for retrieval, synthesis and validation, leveraging **ReAct** and **Chain-of-Thought** patterns for complex reasoning.
- Fine-tuned** Mistral and Llama models using **QLoRA** for medical domain adaptation and established an **RLHF** pipeline with physician validated preference data to align model outputs with clinical communication standards, improving response safety scores by 42%.
- Implemented** a hybrid retrieval architecture combining **Pinecone** vector embeddings for semantic search over unstructured medical literature and **Neo4j** knowledge graphs for structured entity relationships, enabling 65% accuracy on multi-hop reasoning tasks.
- Evaluated on **RAGAS**, achieving 78% answer faithfulness. Performed iterative **prompt engineering** and implemented **constitutional AI** guardrails, blocking 97% of harmful medical content. Incorporated fact-checking agents to prevent hallucinations.

Infosys | Specialist Programmer - Machine Learning Sep 2021 - Jan 2023
Python, Hugging Face, FAISS, MLflow, ETL, Docker, Kubernetes, AWS (EKS, S3, Glue, RDS), PostgreSQL Bangalore, India

- Developed key components of a document intelligence platform for Fortune 500 clients, including an **ETL pipeline** to ingest and process over 100K unstructured legal contracts daily for subsequent information extraction and semantic search.
- Implemented a BERT-based **Named Entity Recognition (NER)** model by fine-tuning a **Legal-BERT** variant using **Hugging Face Transformers**, achieving a 91% F1-score and reducing manual contract review time by 50%.
- Engineered a **semantic search** engine using SentenceTransformers and a **FAISS** index of 10M+ embeddings, utilizing an optimized IndexIVFPQ for approximate nearest neighbor search. This achieved 92% precision@5 with sub-200ms latency.
- Drove model improvements through a rigorous **MLOps** lifecycle with **MLflow** and **Kubeflow**, where data-driven **A/B testing** proved new models were 35% more accurate and proactive drift detection prevented 15 potential production incidents.

VIGIL Lab, IIT Hyderabad | Research Intern - Computer Vision Mar 2021 - Aug 2021
PyTorch, TensorFlow, Object Detection, OpenCV, Git, Weights and Biases Hyderabad, India

- Developed and trained a custom **YOLOv4-OB** model in **PyTorch** for oriented **object detection**, achieving a 12% mAP increase on the VisDrone benchmark and reducing memory footprint by 30% using **network pruning** and **quantization**.
- Spearheaded the optimization of the **feature-learning** pipeline, a process that involved benchmarking EfficientNet backbones, **attention** modules, and **dilation networks** across **PyTorch** and **TensorFlow** to find optimal configurations.
- Established a rigorous experimental pipeline using OpenCV for **data augmentation** to handle diverse aerial conditions and **Weights & Biases** for tracking over 100+ experiment runs, ensuring full reproducibility.

EDUCATION

Northeastern University | MS in AI | Boston, MA - Head TA for NLP | (Previously MS in ECE at CMU) Jan 2023 - May 2025
Relevant Coursework: Machine Learning, Natural Language Processing, Deep Learning, Reinforcement Learning, LLMs, Generative AI, Software Engineering GPA: 3.96/4

NIT Arunachal Pradesh | Bachelor of Technology in ECE | India Aug 2016 - Jun 2020
Relevant Coursework: Software Development, Data Structures, Algorithms, Computer Networking GPA: 3.32/4